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From Individual Balance Control to Collective Dynamics in Ultra-Dense Crowds

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Pedestrian models

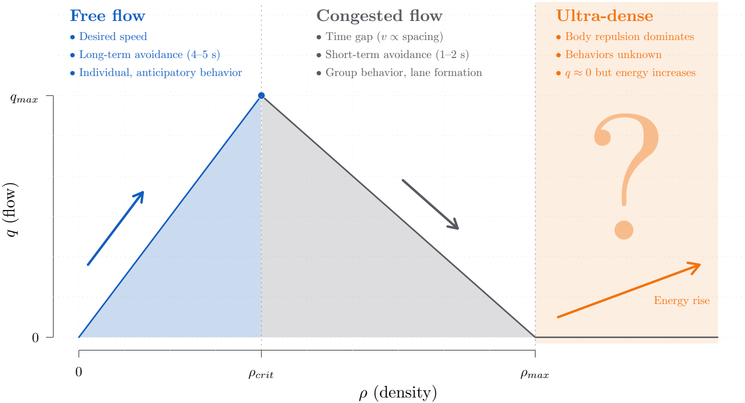
- **Macroscopic models**
  - Pedestrian flow as a continuum (Euler coordinates)
  - First and second order non-linear PDE systems (Hughes model, Eikonal equation, 2nd order models, mean-field games, etc.)

  → Models based on **conservation law**, **fundamental diagram**, **external navigation map**

- **Microscopic models**
  - Individual pedestrian dynamics in Lagrange coordinates (multi-agent system)
  - Coupled non-linear ODE systems – Velocity and force based models (SFM, GCFM, ORCA, ...)

  → Models based **social forces**, **collision-avoidance techniques** or **trajectory optimisation processes**

Density levels



- **Classical modelling concepts of fundamental diagram, navigation map, social force, collision avoidance or trajectory optimisation no longer hold in ultra-dense situations**
- **Dynamics mainly governed by contacts, pushing, body repulsion and biomechanics**

Overview

Ultra-dense crowd: Empirics

Ultra-dense crowd models: State-of-the-art

Two-level pedestrian model

Summary

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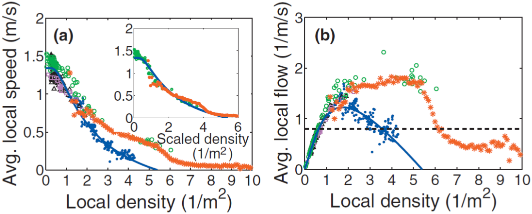
Crowd wave and “turbulent flow” in Makkah (Saudi Arabia, 2006)

Helbing & al. Dynamics of crowd disasters: An empirical study. *Phys Rev E* **75**:046109, 2007

- Videos: youtube/F6EJnMbyM-M, youtube/muKC5bZezlo



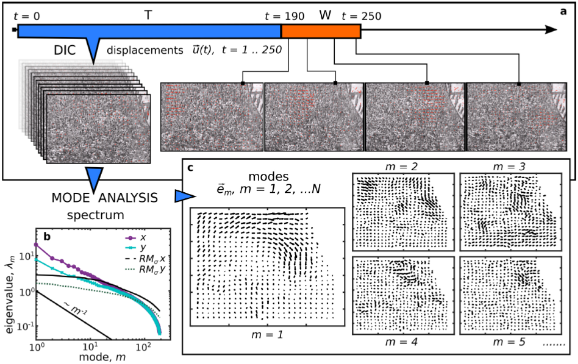
- Fundamental diagram concept no longer holds in extremely dense situation:

“This formula [the fundamental diagram] *is often used as a basis for the dimensioning and design of pedestrian facilities, for safety and evacuation studies. Our video analysis of the Muslim pilgrimage in Mina/Makkah shows, however, that this description needs corrections at extreme densities. In particular, it does not allow one to understand the “turbulent” dynamics causing serious trampling accidents in dense crowds trying to move forward.”*

Pedestrian waves at Oasis rock concert (UK, 2005)

Bottinelli & Silverberg: Can high-density human collective motion be forecasted by spatio-temporal fluctuations? arXiv:1809.07875

- Videos: youtube/BgpdmAtbhbE, youtube/1tPvxBnsSKQ

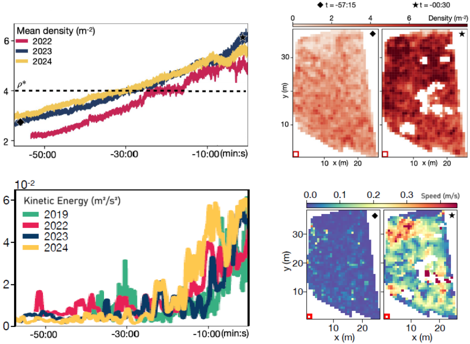


"Generalized model-based approach for real-time monitoring/risk-assessment of crowd collective motion remains an open problem. Here, we take a model-free approach to crowd analysis and show that emergent collective motion can be forecasted directly from video data"

Oscillatory dynamics (San Fermín festival, Love Parade in Duisburg)

Gu & al. Emergence of collective oscillations in massive human crowds. *Nature* **638**:112, 2025

- Videos: nature.com/articles/s41586-024-08514-6#Sec38 (San Fermín festival)
  youtube/QpzISdBE3dA (Love Festival 2010 in Duisburg)



"Collective oscillations in dense crowds arise from a non-reciprocal phase transition towards a chiral oscillatory state. Not only does this description remain consistent across four occurrences of the San Fermín festival but also it holds for the 2010 Love Parade disaster."

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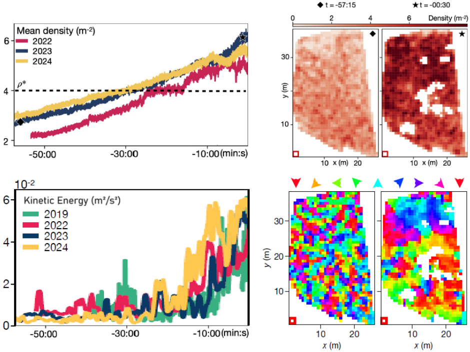

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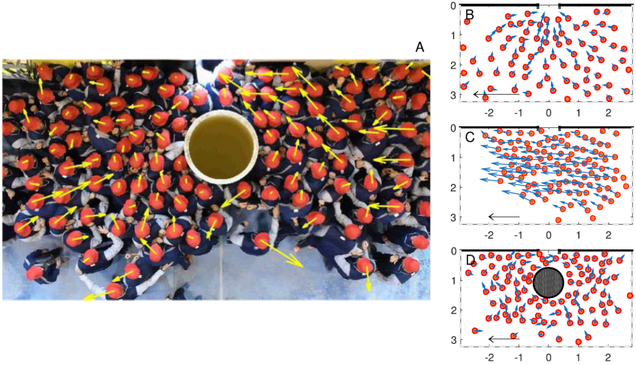
Transversal rushes at bottleneck in case of evacuation

Garcimartin & al. Redefining the role of obstacles in pedestrian evacuation. *New J Phys* **20**:123025, 2018

- Videos: https://iopscience.iop.org/article/10.1088/1367-2630/aaf4ca



"[...] these transversal waves are more likely to occur in the first half of the evacuation (when there is still a large number of people in the room). This feature should be further studied but points towards the importance of the group size in determining the collective behaviour emerging in these kind of complex systems."

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Dense pedestrian dynamics models

- **Microscopic approaches** Pairwise contact forces

  2000 Simulating dynamical features of escape panic. Helbing & al. *Nature* **407**:487
  Body contact and sliding friction forces

  2007 Modeling crowd turbulence by many-particle simulations. Yu & Johansson *Phys Rev E* **76**:046105 – Nonlinear contact forces

  2013 Collective Motion of Humans in Mosh and Circle Pits at Heavy Metal Concerts. Silverberg & al. *Phys Rev Lett* **110**:228701 – Vicsek-like model with algebraic body contact and passive/active pedestrians. Online simulation module

  2015 Velocity-based modeling of physical interactions in dense crowds. Kim & al. *Vis Comput* **31**:541 – ORCA collision avoidance model with contact forces

  2016 Emergent structural mechanisms for high-density collective motion inspired by human crowds. Bottinelli & al. *Phys Rev Lett* **117**:228301, 2016 – Algebraic body contact force

- **Hybrid approaches** Kernel-based continuum models

  2009 Aggregate Dynamics for Dense Crowd Simulation. Narain & al. *ACM Trans Graph* **28**:122 – Contact through incompressibility constraint

  2014 Continuum modeling of crowd turbulence. Golas & al. *Phys Rev E* **90**:42816, 2014
  Frictional body contact force

  2021 SPH crowds: Agent-based crowd simulation up to extreme densities using fluid dynamics. Van Toll & al. *Comput Graph* **98**:306 – Smoothed particle hydrodynamics with collision avoidance and contact force

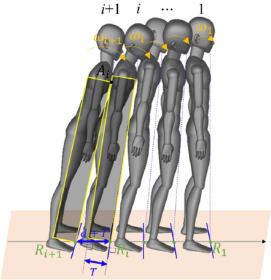
- **Add of contact forces in 2D frameworks**

Modeling human domino process based on interactions among individuals for understanding crowd disasters

Wang & al. *Physica A* **531**: 12178, 2019

"The current models usually consider the movement of the pedestrians in the 2D plane, even though motion in a vertical direction can be simulated in some cases. Therefore, the human body is simplified as an elastic circle or ellipse or as a combination of two shapes [19–21].

From the perspective of biomechanics, however, the movement of pedestrians will be accompanied by changes in body postures and center of gravity. The simplest case can be described as an inverted pendulum model, which is often used to simulate human stepping and to calculate the contact force with the ground [22,23]. This model reveals an important fact that the movement of pedestrians is the result of the three-dimensional (3D) actions rather than 2D actions."



"In dense crowds, the local interactions among pedestrians can easily affect the surrounding individuals because of the small space, which may lead to a cascade effect or domino effect in the crowd. [...] That is, the traditional 2D models cannot accurately simulate these interactions among pedestrians in dense crowds."

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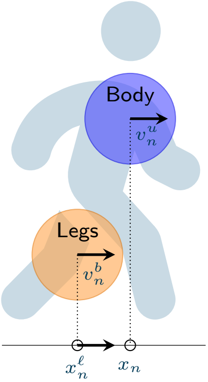


$$\begin{cases} \dot{x}_n = v_n \\ \dot{x}_n^\ell = v_n^\ell \\ \dot{v}_n = \underbrace{\lambda_u (v_n^u - v_n)}_{\text{Unbalancing}} - \underbrace{\lambda v_n}_{\text{Damping}} - \underbrace{\sum_{m \neq n} \nabla V(x_n - x_m)}_{\text{Upper body interaction}} \\ \dot{v}_n^\ell = \underbrace{\lambda_b (v_n^b - v_n^\ell)}_{\text{Balance recovery}} - \underbrace{\sum_{m \neq n} \nabla V_\ell (x_n^\ell - x_m^\ell)}_{\text{Legs interaction}} \end{cases}$$

Parameters:

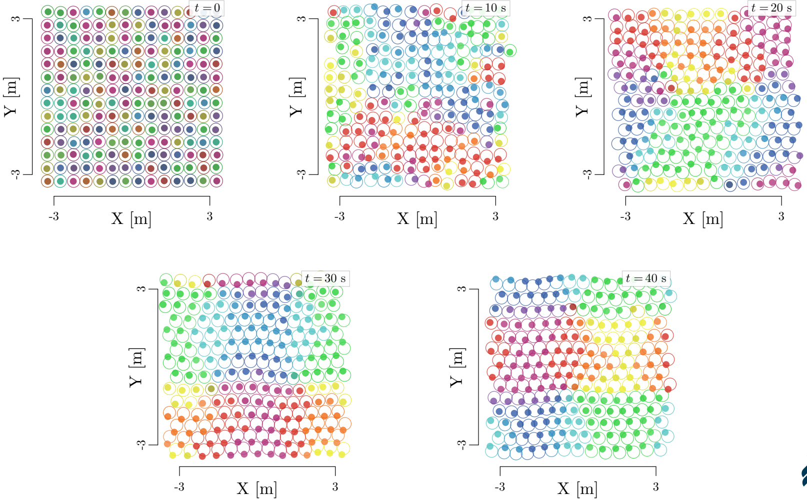
$v^b,\ v^u$	Balancing/unbalancing speed
$\lambda_b,\ \lambda_u$	Balancing/unbalancing rates
$V,\ V_\ell$	Body/Leg repulsion potentials
λ	Damping rate

Online simulation platform

Online simulation module: `antoinetordeux.github.io/Two-Level-Pedestrian-Model`

Simulation results

Large large balancing speed and rate: **Density wave pattern** $\lambda_b = 1\ \mathrm{s}^{-1}$, $\lambda_u = 0.5\ \mathrm{s}^{-1}$, $|v^b| = |v^u| = 1$ m/s



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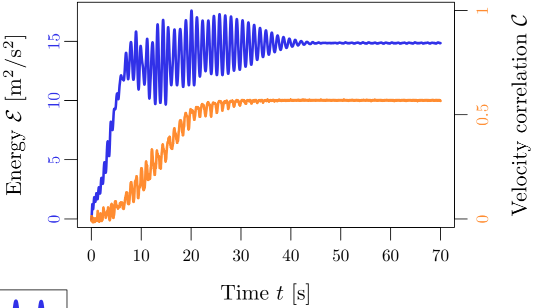
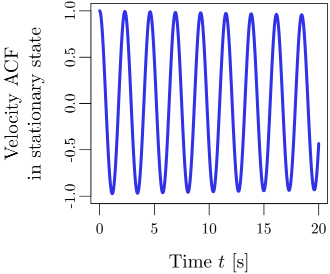
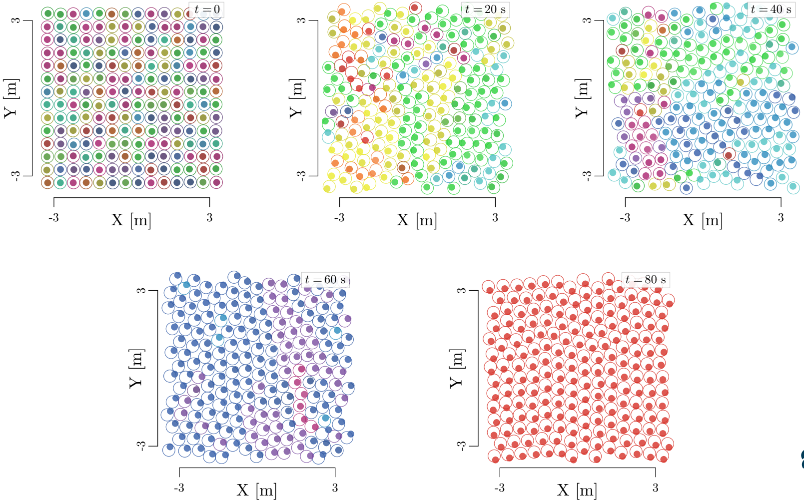




Simulation results

Large unbalancing rate: **Chiral rotation** $\lambda_b = 0.5\ \mathrm{s}^{-1}$, $\lambda_u = 1\ \mathrm{s}^{-1}$, $|v^b| = |v^u| = 0.2$ m/s



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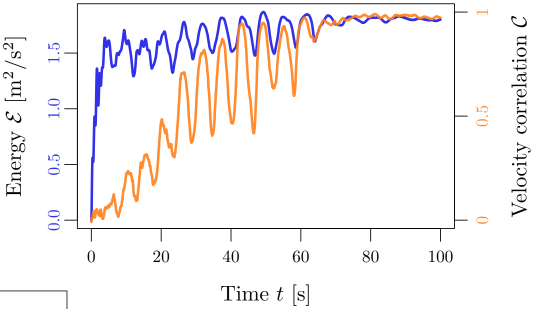
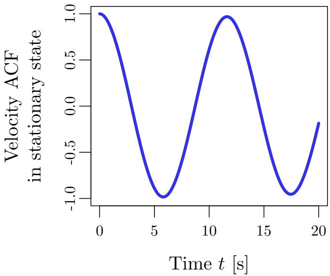
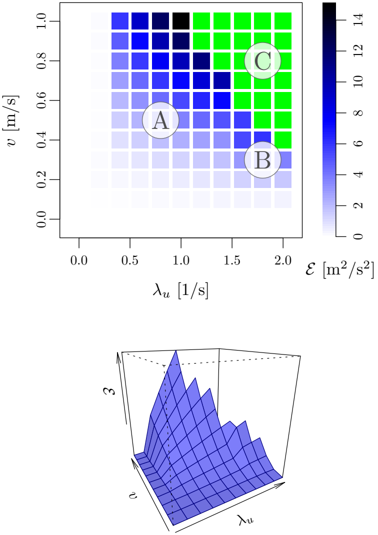




Phase diagram

Energy $\mathcal{E} = \sum_{n=1}^{N} |\mathbf{v}_n|^2$, Velocity correlation $\mathcal{C} = \frac{1}{N} \sum_{n=1}^{N} \frac{1}{N_n} \sum_{m \in D_n} \frac{\mathbf{v}_n \cdot \mathbf{v}_m}{|\mathbf{v}_n||\mathbf{v}_m|}$



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- Minimal **two-level pedestrian model** (legs + upper body) for ultra-dense crowds that captures the basics of **balance–unbalance dynamics**

Main results

- Reproduces **coexisting collective modes** of density waves and chiral rotations
- Emergence from **local interactions only** (no fine-tuning, no global rules)
- Reveals **robust dynamical regimes** with clear phase boundaries

Key insight

- Crowd $\approx$ network of **coupled inverted pendulums**
- **Biomechanical balance control** is the dominant organizing principle

$\rightarrow$ Details of contact forces are **secondary**

Outlook

- Extend model: nonlinearities, heterogeneity, friction, falling
- **Calibration & validation** with real crowd data

$\rightarrow$ Toward **predictive tools for crowd safety**

Thank you for your attention !

Download slides

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